

Ryder Dutil

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Education

University of Nevada, Reno

Expected Spring 2028

- Bachelor of Science in Aerospace Engineering
- Bachelor of Science in Mechanical Engineering

Projects

Analysis of an Electromechanical System (Rail Gun)

Team Project | University of Nevada, Reno | April 2026 - May 2026

- Derived transfer functions for a linear DC machine using **Laplace transforms** and **Cramer's rule** to model a rail gun system
- Analyzed step responses, pole characteristics, and steady-state dynamic behavior using Final Value Theorem
- Generated Bode plots and Simulink block diagram to verify analytical results against simulation
- Applied **MIMO transfer function** framework to evaluate system performance under varying voltage and load conditions

ADA-Compliant Ramp Design and Prototyping for Campus Facility

Team Project | University of Nevada, Reno | Jan. 2026 - May 2026

- Led design of a Nevada ADA-compliant ramp as primary SolidWorks modeler, applying self-researched state accessibility standards to drive geometric and compliance decisions
- Interpreted Nevada accessibility regulations to ensure proper slope, landing dimensions, width, and edge protection
- **Directed** a four-person team as lead designer, coordinating geometry iterations, constraint validation, and constructibility
- Produced a 3D-printed prototype and formally presented design rationale and compliance justification
- Conducted **full lifecycle** cost analysis from construction to teardown, including material volume calculations through **SolidWorks** and local market pricing research

Balanced Six-Sided Die Assembly in SolidWorks

Individual Project | University of Nevada, Reno | Oct. 2025 - Dec. 2025

- Designed a balanced six-sided die across six distinct materials, achieving CG error of 0.115% (req. <2%), MOI error of 1.495% (req. <3%), and POI error of 0.122% (req. <0.2%)
- Developed a custom **MATLAB** solver using multi-start optimization to sweep discrete thickness combinations and simultaneously satisfy center-of-mass, moment of inertia, and products of inertia tolerances
- Produced a full SolidWorks drawing package including individual part drawings for each face, exploded assembly view, and BOM

Experience

Space Ocean Corporation — Assistant Deputy Director of Engineering

Remote | October 2025 - PRESENT

- Led development of program review materials and presented to the CEO & design review team in support of spacecraft Instrument Baseline Design Review (**IBDR**)
- Directed and collaborated with engineers across the organization on slide deliverables and documentation requirements in support of IBDR preparation
- Developed **technology statements** & **technical briefs** for internal teams and external partners
- Engaged with vendors on component specifications and interfaced with university and external technology partners through technical meetings and ongoing discussions
- Created and iterated **SolidWorks models** to support early-stage design concepts and investor demonstrations
- Authored and reviewed program documentation including risk assessments, mission statements, **CONOPS**, and **SRD** to maintain accuracy and configuration control

Baskin-Robbins — Shift Lead / Opener

Reno, NV | Sept 2022 - PRESENT

- Lead 1-3 associates per shift to ensure efficient operations and customer service
- Open and close store, reconcile cash drawer, and secure funds in safe
- Maintain high standards of cleanliness and organization store-wide

Skills

Technical

SolidWorks (CSWA Certified) | **GD&T** | **MATLAB**
NI Multisim | Microsoft Office Suite, to include Microsoft Excel | Soldering (through-hole & surface-mount)

Soft

Detail Oriented | Problem Solving and Critical Thinking | Communication | Teamwork | Time Management | Adaptive & Flexible | Leadership | Collaboration | Professional Writing

